ZOO 501 Mid Current Solved

What is gastrulation?

Gastrulation is a phase early in the embryonic development of most animals, during which the single-layered blastula is reorganized into a multilayered structure known as the gastrula. **(Internet)**

Who discover sperm?

A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in 1678

Name the three germ layer in animals?

1. Ectoderm 2. Mesoderm 3. Endoderm

How many type of mRNA are present in sea urching egg?

sea urchins contain 25,000 to 50,000 different types of mRNA.

Differentiate B/W Promoter and Enchancer?

Promoter: A region, which is responsible for the binding of RNA polymerase and for the subsequent initiation of transcription.

Enhancer: It is a DNA sequence that can activate the utilization of a promoter, controlling the efficiency and rate of transcription from that particular promoter. The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs

What are Somites?

Somites are precursor populations of cells that give rise to important structures associated with the vertebrate body plan and will eventually differentiate into dermis, skeletal muscle, cartilage, tendons, and vertebrae. ... Formation begins as paraxial mesoderm cells organize into whorls of cells called somitomeres. **(Internet)**

What is different B/W Sex and Fertilization?

Fertilization: Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

Sex involves the combining of genes from two different individuals into new arrangements.

What are the event of Fertilization?

Although the details of fertilization vary from species to species, conception generally consists of four major events:

- Contact and recognition between sperm and egg.¬ In most cases, this ensures that the sperm and egg are of the same species.
- Regulation of sperm entry into the egg. Only one sperm can ultimately fertilize the egg
- Fusion of the genetic material of sperm and egg.
- Activation of egg metabolism to start development.

What are two main startegie of developmental commitment?

There appear to be two major strategies for establishing commitment and hence initiating the series of events that result in cell differentiation.

The inheritance of cytoplasmic determinants: Cytoplasmic determinants are the molecules in cytoplasm that can help to determine cell fate. The asymmetric distribution of cytoplasmic determinants indicates that the mechanism of differentiation is entirely intrinsic

The perception of external inductive signals: The process where one cell or group of cells changes developmental fate of another is termed induction. It is extrinsic process that depends on the position of a cell in the embryo.

How Egg is protected form sunlight? Which of Egg protected form sunlight?

Many eggs contain ultraviolet filters and DNA repair enzymes that protect them from sunlight; some eggs contain molecules that potential predators find distasteful; and the yolk of bird eggs even contains antibodies.

Composition of Egg Yolk? Explain composition of Egg Yolk?

Yolk vitamins and minerals. **Yolks** hold more than 90% of the calcium, iron, phosphorus, zinc, thiamine, vitamin B_6, folate, vitamin B_{12}, and pantothenic acid of the **egg**. In addition, **yolks** cover all of the fat-soluble vitamins: A, D, E, and K in the**egg**, as well as all of the essential fatty acids. **(Inernet)**

Conditional Specification?

Conditional specification: o Characteristic of all vertebrates and few invertebrates. o Specification by interactions between cells. Relative positions are important. Variable cleavages produce no invariant fate assignments to cells. o Massive cell rearrangements and migrations precede or accompany specification. o Capacity for "regulative" development: allows cells to acquire different functions.

What is difference B/W protocadherin and other Cadherin?

Protocadherins: Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they lack connections to the cytoskeleton through catenins. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during Xenopus gastrulation.

E-cadherin: Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1-cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.
P-cadherin: Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.
N-cadherin: Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their Ecadherin expression. It is also highly expressed on the cells of the developing central nervous system.

Describe anatomical approachesof developmental biology?

The basis of all research in developmental biology is the changing anatomy of the organism. What parts of the embryo form the heart? How do the tissues that form the bird wing relate to the tissues that form the fish fin or the human hand? Following important parameters weave together to form the anatomical approaches to development. Comparative embryology¬ Evolutionary embryology¬ Teratology¬ Mathematical modeling¬

What is Microvilli? Microfilaments?

Microfilaments are necessary for cell division, and they also are used to extend the egg surface into small projections called microvilli, which may aid sperm entry into the cell.

Difference B/W Somatic Cell and Germ Cell?

Germ Cell: In many species a specialized portion of egg cytoplasm gives rise to cells that are the precursors of the gametes (the sperm and egg). The gametes and their precursor cells are collectively called germ cells, and they are set aside for reproductive function.
Somatic Cell: All the other cells of the body are called Somatic cells. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the formation of a new generation) is often one of the first differentiations to occur during animal development.

How the gene activity of human Is obtain?

Knowledge of gene activity in humans can be obtained by candidate gene mapping or positional cloning.

Transcription Factor Domain?

Transcription factors have three major domains. 1. DNA-binding domain 2. Protein-protein interaction domain 3. Trans-activating domain.

Define adherence Junction?

Adherens junctions (AJs) are cell-cell adhesion complexes that are continuously assembled and disassembled, allowing cells within a tissue to respond to forces, biochemical signals and structural changes in their microenvironment. **(Internet)**

What is function of vitelline envelop?

The vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

Define Organogenesis in term of germ layer?

Once the three germ layers are established, the cells interact with one another and rearrange themselves to produce tissues and organs. This process is called organogenesis. Many organs contain cells from more than one germ layer.

Sperm Egg reorganization?

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Development of hierarchy of animals?
The developmental hierarchy is the hierarchical series of decisions involving specific pre-existing cell types. These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate.

Structure of Sperm?
The **sperm** head. The head is oval-shaped and is only 10% of the overall length of the **sperm**. It contains the male genetic material in a **structure** called the nucleus. ... In addition to DNA, the **sperm** head also contains a**structure** called the acrosome. **(Internet)**

Mosaic development?
Autonomous specification gives rise to a pattern of development referred to as mosaic development, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts.

Capacition?
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Poly A tale Function? Role of Poly A tale in mRNA?
The poly(A) tail Confers stability on the messenger RNA.υ Allows the mRNA to exit the nucleus.υ Permits the mRNA to be translated into protein.

Vegetal Pole?
The yolk-rich pole is referred to as the vegetal pole; the yolk concentration in the animal pole is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

What is the larve stage of frog?
A tadpole (also called a pollywog) is the **larval stage**in the **life cycle** of an amphibian, particularly that of a**frog** or toad. They are usually wholly aquatic, though some species have tadpoles that are terrestrial. **(Internet)**

How genetic material exchange in paramecium?
The micronucleus is the generative, or germline nucleus, containing the **genetic material** that is passed along from one generation to the next. **Paramecium** reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis. **(Internet)**

Acrosome and Its role?
Acrosome: In Eutherian mammals the **acrosome** contains digestive enzymes (including hyaluronidase and acrosin). These enzymes break down the outer membrane of the ovum, called the zona pellucida, allowing the haploid nucleus in the sperm cell to join with the haploid nucleus in the ovum. **(Internet)**

Protostome?
(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location. There are two major branches of the protostomes. **Ecdysozoa, Lophotrochozoa.**

Cadherin types?
E-cadherin P-cadherin¬ N-cadherin¬ Protocadherins¬

Role of Bicoid and Nanos in synctial specification?
The Bicoid and Nanos proteins form a coordinate system based on their ratios, such that each region of the embryo will be distinguished by a different ratio of the two proteins. Those nuclei in regions containing high amounts of Bicoid and little Nanos will be instructed to activate those genes necessary for producing the head. Those nuclei in regions that have little or no Bicoid and plenty of Nanos will be instructed to form the abdominal Structures. Insects such as Drosophila use all three modes of specification to commit their cells to particular fates.

What are the function of zone pellicuda?
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Blastcoel and Its function?
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